

Warrior Trading *Assistant*

A personal desktop assistant for applying Ross Cameron's Warrior day-trading strategy in real time — what it actually does today, how each screen works, and exactly how it decides what to recommend.

▲ Long-only

Small-cap momentum · \$2-\$20

It proposes — you execute

Reduce risk first

How to read this guide. It describes **what is actually built today**, not the eventual vision. Where something is planned-but-not-wired, it says so plainly (see §6.2). The authoritative engineering reference is `CLAUDE.md` ; this is the human-facing companion to it. The colors below carry meaning throughout:

- permitted / go / built
- blocked / hard stop / risk
- a caution or value that matters
- a tool or context

Orientation

What the program is for, the hard constraints baked into it, and — most importantly — an honest map of what is built today versus what is deliberately deferred.

1.1 — THE THESIS

What This Program Is

A personal desktop assistant for applying **Ross Cameron's Warrior day-trading strategy** while you trade. It watches the market in real time, finds the kind of volatile small-cap momentum stocks Ross trades, checks each potential trade against his risk rules, and surfaces **buy/sell recommendations and exit management** on screen.

Two principles shape everything it does:

1 · Reduce risk first

Every design choice favors protecting capital over chasing upside, while still generally following Ross's strategy.

2 · Be easy to use

One native window, plain-language recommendations, and a single most-important number always in view.

Hard constraints baked into the program

CONSTRAINT · LONG-ONLY

It only ever recommends **buying** (then selling to exit). Bear patterns — double tops, head-and-shoulders, breakdowns — are used **only as exit signals**, never as entries.

CONSTRAINT · IT PROPOSES, YOU EXECUTE

The program **never places orders**. It recommends; you decide and place the trade in your own broker, then tell the program what you did (or, with IBKR, it reads your fills live). A deliberate safety design — **you are always the one pulling the trigger**.

CONSTRAINT · SMALL-CAP MOMENTUM, SHORT HOLDS

\$2–\$20 stocks, up big on the day, low float, high relative volume — held **seconds to a few minutes**.

1.2 — READ THIS FIRST

What's Built vs. What Isn't

The single most important thing to understand before trusting the program: the recommendations come from a **deterministic rules engine** today — not from Claude reasoning live over the data. That brain layer is parked. Here is the honest map.

✓ BUILT AND WORKING

- Per-user **config layer** (your risk anchors)
- Two **P&L providers**: Manual entry & **IBKR** paper trading (TWS API)
- **Scanner** — Five-Pillars candidate scan (Massive), true **float** via Ortex, **country** via SEC EDGAR
- **~18 entry-setup detectors** (micro pullback, bull flag, ABCD, VWAP reclaim, PM high/pivot breaks, dip trade...)
- **Hard risk gates + cautionary flags**
- **Exit engine** (scale-out + B/E stop, every-tick catastrophic stop, loss-cap/goal, MACD flip, MA break, bear-pattern exits)
- **Position sizing** (risk-based, regime-aware)
- **Market-regime** classification
- The **desktop UI** (dashboard, scanner board, recs feed, exits panel, ticker tape)
- An offline **backtest harness**

X NOT YET BUILT / DEFERRED

- The **Claude "brain"** — recs are from the rules engine, not live Claude reasoning. Parked.
- **Sub-second print-trigger execution** — the live tick feed isn't wired, so the engine runs in **bar-close mode** (1-min closes). The every-tick catastrophic stop is the exception.
- **Hijack-state & rehab-state gates** — in doctrine but not live; your **broker-side caps are the substitute**.
- **News pillar** — disabled (no vendor wired). The four price/volume/float pillars carry the scan.
- **Level 2 / depth-of-market** — a few setups lean on tape in doctrine; the app runs them on bar-structure proxies for now.



EXITS ARE THE LEAST LIVE-VALIDATED SUBSYSTEM

They're unit-tested and internally consistent, but have had the least real-market exposure. Watch them closely on your first live mornings.

Getting Started

One-time setup, how to launch, and the single decision that changes how the program tracks your money: Manual entry versus a live Interactive Brokers connection.

2.1 — ONE-TIME SETUP

Prerequisites & Setup

You need a Windows 11 machine, Python 3.12, and the `uv` package manager. Market data comes from Massive; Ortex and IBKR are optional.

Required

Windows 11, Python 3.12, `uv`, and a **Massive** (formerly Polygon) Stocks Advanced data subscription.

Optional

Ortex for true float, and **Interactive Brokers** for live P&L / positions. Without them the scan still runs on four pillars and you use the Manual provider.

Install dependencies

```
# from the project root
uv sync                    # install dependencies into .venv
```

Create your config

1 Copy `config/schema.example.yaml` → `config/user.yaml` . **That file is gitignored** — your API keys never get committed.

2 Fill in your **anchors** — the values Ross's ratio-based rules need (full position size, daily goal, max daily loss, your "barely-feel-it" per-trade loss, experience tier). See the table below.

3 Fill in vendor keys: `massive.api_key` , and optionally `ortex.api_key` .

4 If using IBKR, set the `ibkr` section (host / port / client_id). **Port 7497 is paper trading**; 7496 is live. Use paper until you trust it.

The anchor values

ANCHOR	WHAT IT IS
<code>full_position_size_shares</code>	Your full position size (e.g. 1000).
<code>daily_profit_goal_dollars</code>	Your daily goal (e.g. 2000).
<code>max_daily_loss_dollars</code>	Your hard daily stop (e.g. 3000).
<code>personal_barely_feel_loss_dollars</code>	The per-trade loss you'd "barely feel" — drives risk-based sizing.
<code>user_experience_tier</code>	Your experience tier (modulates sizing quality).



THE SCHEMA IS THE SOURCE OF TRUTH

`config/schema.example.yaml` is authoritative for **exact field names and which are required vs. optional**. If a field name in this guide differs from the schema, trust the schema.



HOT RELOAD

Edits to `config/user.yaml` while the app is running are picked up **without a restart**. A failed reload keeps the previous config and logs a warning.

2.2 - BOOT

Launching the Program

```
uv run python -m app.main
```

A native desktop window opens.



ALWAYS LAUNCH VIA UV RUN

A bare `python` lacks the timezone data the app needs and will report spurious errors (`ZoneInfoNotFound`).

Health checks you can run without the GUI

```
uv run python -u scripts/ibkr_smoke_test.py      # verify the TWS connection
uv run python -u scripts/runtime_smoke_test.py    # boot the scan engine headless
```

2.3 - HOW P&L IS TRACKED

Manual vs. IBKR

The program tracks your positions and day P&L through one of two providers. Selection is **automatic from your config**: a real

`ibkr.account_id` (not the `REPLACE_WITH...` placeholder) → IBKR; otherwise → Manual.

	MANUAL	IBKR
Positions / P&L source	what you type in	live from TWS/Gateway
"Record Fills" card	shown — you log entries/exits	hidden — fills flow automatically
Requires TWS running	no	yes (paper = port 7497)
Banner	—	shows the connected account; warns on disconnect

Use Manual...

to try the program with no broker attached. You log your own fills so the exit engine can manage positions.

Use IBKR (paper)...

for a realistic dry run where the program sees your real (simulated) fills automatically.

The Interface

One window, six stacked panels — from the cosmetic crawl at the very top to the panel where you log your fills.

3.1 — TOP TO BOTTOM

The Screen, Panel by Panel

When the window is open you'll see six panels, stacked top to bottom. The Session Risk dashboard is the one to keep your eyes on.

1 **Ticker tape** (very top) — a decorative news-style crawl of **every NASDAQ-listed common stock, alphabetical**, with live price and % change. Purely cosmetic; it does not drive any decision.

2 **Session Risk dashboard** — your at-a-glance safety panel (detailed below).

3 **Scanner board** — the autonomous candidate scan. Columns: ticker, country (China/Hong Kong flagged elevated-risk), last price, % change, RVOL, float, verdict. It **refreshes itself** on a timer — there is no refresh button.

4 **Recommendations feed** — live trade recommendations as setups fire (detailed below).

5 **Position Management (exits) panel** — while you hold, the exit engine surfaces management signals here, sorted by **urgency**. IMMEDIATE ones (stops) alarm.

6 **Record Fills card** (Manual provider only) — log an entry, an exit, or a mark-to-market price so the program can track P&L and manage exits.

Inside the Session Risk dashboard

Day-P&L-vs-max-loss meter

How much of your daily loss cap you've used (amber ~70%, red ~90–100%). **The single most important number on the screen.**

Goal meter

Progress toward your daily profit goal.

Regime chip

The market-regime classification (cold / lukewarm / hot), which tightens or loosens sizing and caution.

Market-shape flags & banners

Round-trip-prone (leaders giving back gains — favor exits), *dispersed attention* (too many movers), *rehab* (reserved).
Banners: the **daily-loss-cap padlock**, **goal-hit**, and stand-aside notices.

Recommendation states

Each card in the feed shows the **setup name**, **entry / stop / target**, the pullback number, any **cautions** (e.g. "below VWAP", "MACD bearish on 1m"), and a **recommended share count** with rationale. Its status is one of:

STATUS	MEANING
PERMITTED	Every gate passed; you may take it as a fresh entry.
SCALE_IN	Allowed, but you already hold a partial position — add to it.
SCALE_OUT	Informational ("you're holding the runner").
BLOCKED	A gate failed; the reason is named.
WARMING_UP	A gate can't evaluate yet (data still warming).

Under the Hood

One deterministic pipeline: scan, detect, gate, recommend — then manage every position you hold on every price update. Here's exactly how each stage decides.

4.1 — THE ENGINE

The Pipeline & Gate Stack

The engine runs one deterministic pipeline. A setup is recognized, the gate stack vets it, and only a recommendation that clears every **hard gate** is marked PERMITTED.

- ▶ scan the market → build the tradeable universe (Five Pillars)

- ▶ stream 1-min bars for each candidate

- ▶ detectors recognize a setup → gate stack vets it

- ▶ recommendation appears in the feed

- 🔄 while you hold: every price update → exit engine → exit signals

Hard gates — these block a recommendation outright

HARD GATE • DAILY-LOSS-CAP PADLOCK

Once your day P&L hits `-max_daily_loss`, **no new entry** is permitted for the rest of the session.

HARD GATE · OPEN-POSITION

Don't double-enter a name you already hold (instead it may suggest a scale-in).

HARD GATE · THE FOUR UNIVERSE PILLARS (ALL MUST BE TRUE)

Price \$2–\$20 · **% change** up $\geq 10\%$ on the day · **RVOL** $\geq 5\times$ (today's volume vs. its 50-day average) · **Float** $< 20\text{M}$ shares (true float via Ortex when available).

Cautionary flags — these inform but never block

You see these on the card and decide for yourself:

MACD on the 1-minute

bearish / warming.

Below VWAP

price below, or VWAP warming.

News

no catalyst / unverified — *currently disabled as a pillar.*



WHY MACD / VWAP / NEWS ARE CAUTIONS, NOT GATES

Ross's biggest winners repeatedly trade against a cold 1-min MACD, below VWAP, with no news. Price action + supply/demand + Level 2 are primary; these three are advisory. A deliberate 2026-05-29 design decision — see [CLAUDE.md](#) .

The **daily chart** is used as a *context filter* (overhead resistance, room to run, blue-sky) — **never as an entry trigger**.

4.2 — RECOGNITION

The Entry Setups (Detectors)

All detectors are **long / bullish** recognizers fed by 1-minute bars. When one fires on an armed candidate, it produces a recommendation card for the gate stack to vet.

Micro pullback

Bull flag

ABCD

Break of pivot

Cup & handle

Inverted head-and-shoulders

HOD breakout

Pre-market high break

Pre-market pivot break

Red-to-green

VWAP reclaim

Dip trade

Curling

First pullback

Double bottom

Half / whole-dollar breakout

MA pullback 5-minute



RTH-ANCHORED DETECTORS STAY QUIET PRE-MARKET

Red-to-green and **first-pullback** are anchored to the regular session and intentionally do not fire during pre-market.

4.3 — POSITION MANAGEMENT

The Exit Engine

This is the position-management half. While you hold, every price update runs the exit engine; signals appear in the exits panel sorted by urgency.

EXIT • FIRST-PROFIT SCALE-OUT + BREAK-EVEN STOP

The cleanest hardwired rule: at the first profit target, **sell a portion**, then move the stop on the runner to **break-even**. The two fire together.

EXIT • EVERY-TICK CATASTROPHIC STOP

The hard stop is checked on **every** price update (not a bar close), because these stocks move multiples of the stop width within a single bar. Once break-even is active, this same check enforces the break-even exit.

EXIT • DAILY LOSS-CAP / GOAL

Portfolio-level: escalates toward the cap, locks in at the goal.

EXIT • DISCRETIONARY SIGNALS

MACD flip, 9-EMA / 20-EMA / VWAP break, 50%-retrace give-back, 2-minute breakout-or-bailout, and the **bear-pattern confirmations** (double top, head-and-shoulders, top reversal, LOD break, bear flag) — used here as *exit* signals, consistent with long-only.

4.4 — HOW BIG

Position Sizing

The recommended share count is **risk-based**, and it is **advisory** — it never blocks a trade.

The formula, roughly

Your per-trade "**barely-feel-it**" loss ÷ the **stop distance** — then **capped** at your full position size and **shrunk** by a quality-tier modifier and a cold-market modifier (**×0.70 in a cold regime**).

i **SIZING NEVER GATES**

It informs the recommended share count, but a trade is never blocked on size.

Trading With It

The full loop of a session — and the layered guardrails that stand between you and a bad day, in priority order.

5.1 — THE LOOP

A Session, Start to Finish

- 1 Before the open:** start TWS/Gateway (if using IBKR) and **set your broker-side daily max-loss and max position size** (see §5.2 — do this every day, it's your real backstop). Launch the app. Glance at the dashboard to confirm your anchors loaded.
- 2 Watch** the scanner board fill in and the recommendations feed.
- 3 On a PERMITTED recommendation you like:** place the trade **in your broker** at (or near) the suggested entry, with the suggested stop and size. On the Manual provider, **record the fill** so the exit engine starts managing the position.
- 4 Manage with the exits panel:** take the first-profit scale-out when it fires (and let the runner ride the break-even stop), and act **immediately** on a catastrophic-stop alarm.
- 5 Respect the risk dashboard:** if the loss-cap padlock engages, you're done adding for the day — the program stops permitting new entries.
- 6 Stop when you decide.** The program runs whenever you run it; there's no forced time window.

5.2 — GUARDRAILS

Risk Controls & Guardrails

In priority order — broker-enforced limits first, because in a "hijacked" emotional state a trader overrides in-app warnings.

PRIORITY 1 · BROKER-SIDE CAPS — YOUR PRIMARY DEFENSE

Set a **daily max-loss** and a **max position size** directly in IBKR. **Non-negotiable for live money.** The broker must enforce the hard limits; the in-app padlock is a secondary layer.

PRIORITY 2 · IN-APP DAILY LOSS-CAP PADLOCK

Blocks new entries once the cap is hit.

PRIORITY 3 · SCALE-OUT + BREAK-EVEN STOP

Takes risk off at first profit.

PRIORITY 4 · EVERY-TICK CATASTROPHIC STOP

The hard floor on every position.

PRIORITY 5 · REGIME-AWARE SIZING

Smaller in cold / round-trip-prone markets.



DEFERRED GUARDRAILS

The *trader-rehab* state machine (auto-shrink after red streaks) and a *hijack-state* gate are documented in the doctrine but **not live**. Until they are, your broker-side caps cover that ground.

Tools & Limits

The offline backtest harness, an honest accounting of what is not built, common fixes, and a map of the key files.

6.1 — LOOKING BACK

The Backtest Harness

An offline tool to approximate how the program *would have* performed on past pre-market sessions. It replays historical 1-minute bars through the same detectors, gates, sizing, and exit engine the live app uses.

```
# last 5 trading sessions, 7-9am ET, default 0.20% slippage
uv run python -u scripts/backtest_premarket.py --days 5 --out report.md

# a specific past week (walks back from --asof, which is excluded)
uv run python -u scripts/backtest_premarket.py --days 5 --asof 2026-05-22 --out price.md

# tune the realism / window
uv run python -u scripts/backtest_premarket.py --slippage-bps 35 --window 04:00-09:30
```

Three fill models — so you see a range, not one misleading number

Optimistic

Perfect fills — the upper bound.

Realistic

Tunable slippage on market-style fills — the central estimate (`--slippage-bps` , default 20 = 0.20%).

Conservative

Pessimistic fills — the lower bound.

It also reports **peak capital deployed per day** (the most buying power tied up at once).



IT IS AN APPROXIMATION — READ IT THAT WAY

It cannot model Level 2; true sub-second fills are coarsened to 1-minute bars (so fill prices carry real error — that's why there are three models); and the daily universe is reconstructed from each day's movers (so it can miss pre-market-only runners that faded). Read it as "**roughly green or red, which setups fired, how often stops vs. targets hit**" — not "you would have made exactly \$X."

6.2 — HONEST ACCOUNTING

Limitations & What Is *Not* Built Yet

No real-time Claude brain

Deterministic rules produce the recommendations today.

Bar-close, not sub-second

Entries / most exits evaluate on 1-minute closes (the catastrophic stop is per-tick on the live path); the print-trigger feed isn't wired.

Hijack / rehab gates not live

Rely on broker-side caps.

News pillar disabled

Four pillars carry the scan.

Level 2 proxied

Setups that doctrine says need tape run on bar structure.

Exits least live-validated

Unit-tested and internally consistent, but the least real-market exposure. Watch them closely on your first live mornings.

6.3 — WHEN SOMETHING'S OFF

Troubleshooting & Operations

SYMPTOM	FIX
Spurious <code>ZoneInfoNotFound</code> errors	Always use <code>uv run</code> — bare <code>python</code> lacks <code>tzdata</code> .
Scanner empty / "waiting for first scan"	Check <code>massive.api_key</code> is set and not the placeholder. Off-hours the board shows the prior session's gainers.
IBKR disconnect banner	TWS/Gateway isn't running, or the port/client_id is wrong. The app auto-reconnects when it returns.
Config change not taking effect	A malformed <code>user.yaml</code> is rejected and the previous config kept — check the console for a WARN line.
Verify the install	<code>uv run pytest</code> runs the full test suite.

6.4 — THE MAP

Reference — Key Files

PATH	WHAT IT IS
CLAUDE.md	The authoritative engineering reference & invariants.
config/schema.example.yaml	Config field reference (copy to <code>user.yaml</code>).
app/main.py	Launch entry point & provider picker.
app/runtime.py	The live scan → detect → gate → exit engine.
app/gates/	Hard gates (<code>composite.py</code>) and cautions (<code>cautions.py</code>).
app/detectors/	The entry-setup recognizers.
app/exits/	The exit-trigger roster.
app/sizing/ · app/scanner/ · app/monitor/	Sizing, scan/vendors, exit orchestration.
app/web/	The UI (<code>index.html</code> , <code>main.js</code> , <code>styles.css</code>).
app/backtest/ + scripts/backtest_premarket.py	The backtest harness.
playbook/	The machine-readable distillation of Ross's course (the strategy source-of-truth).
Claude Code Session Handoff Documents/	Cross-session status; read the most recent first.

"Reduce risk first — then generally follow the strategy."

This guide describes the Warrior Trading Assistant as built in Phase 3 (runtime app). It is an explanation of how a personal tool works — not financial advice, not a recommendation, and not a guarantee of any result. The program proposes; you execute. Trading small-cap stocks carries substantial risk of loss. **Last updated: 2026-05-31.**